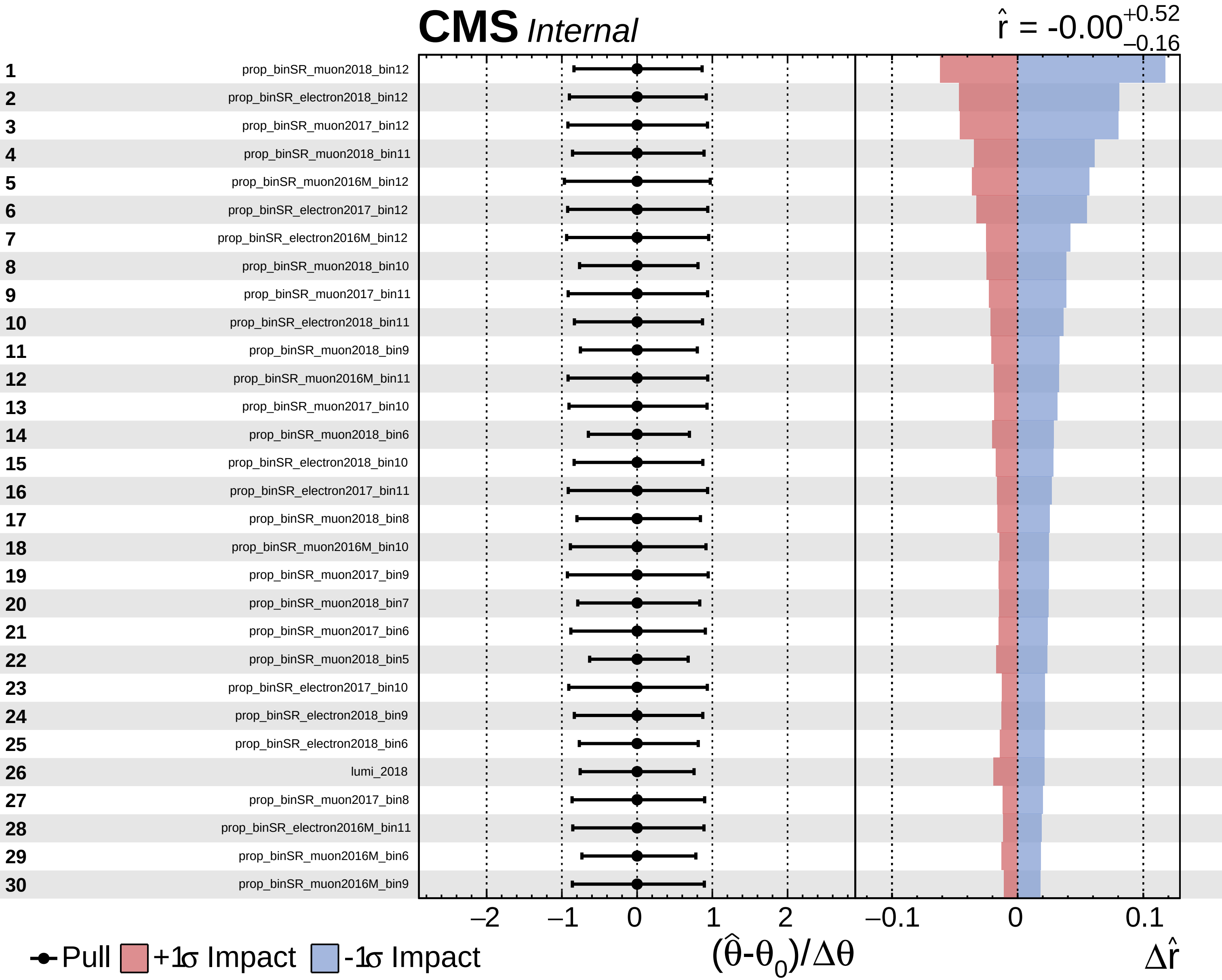
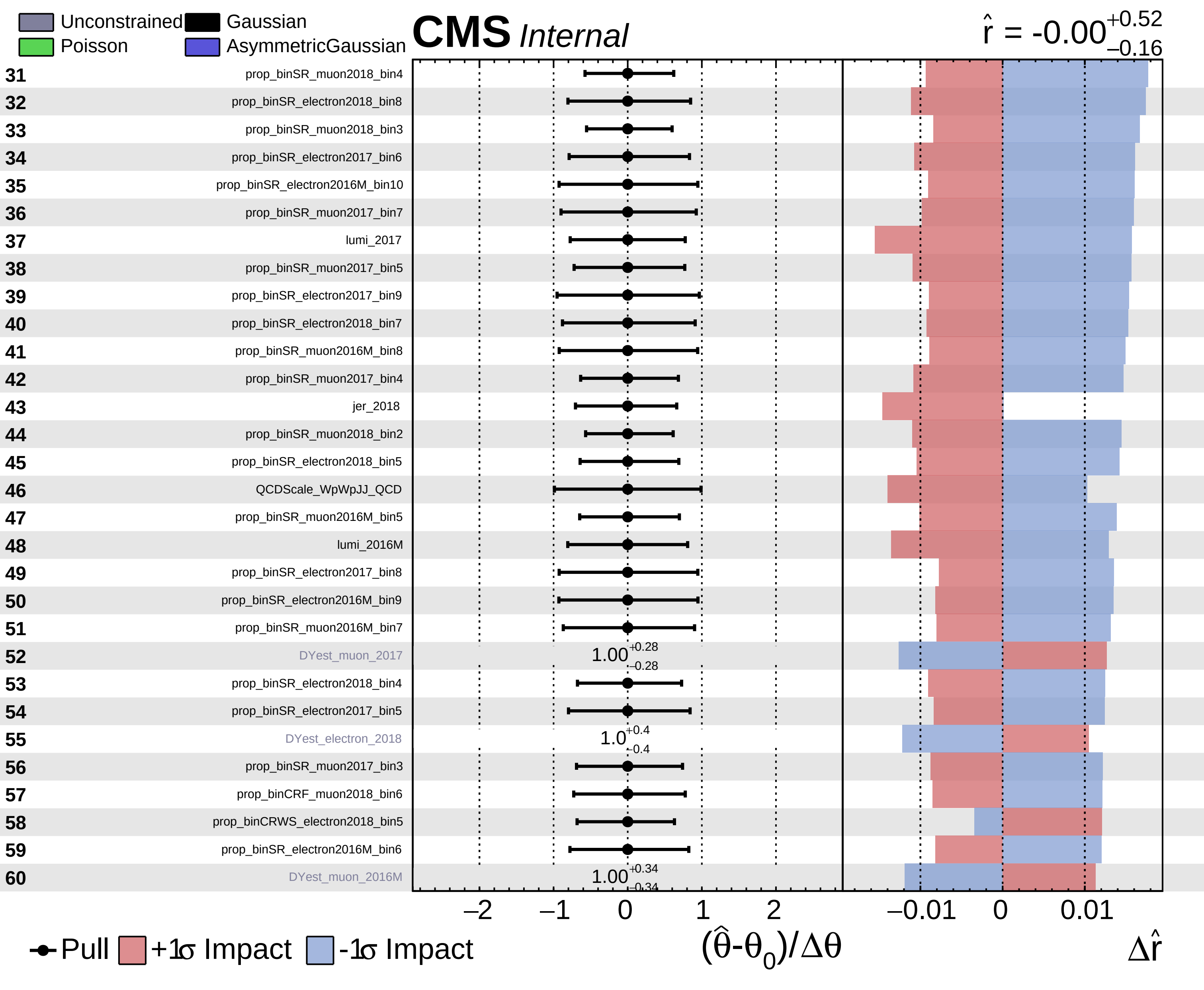
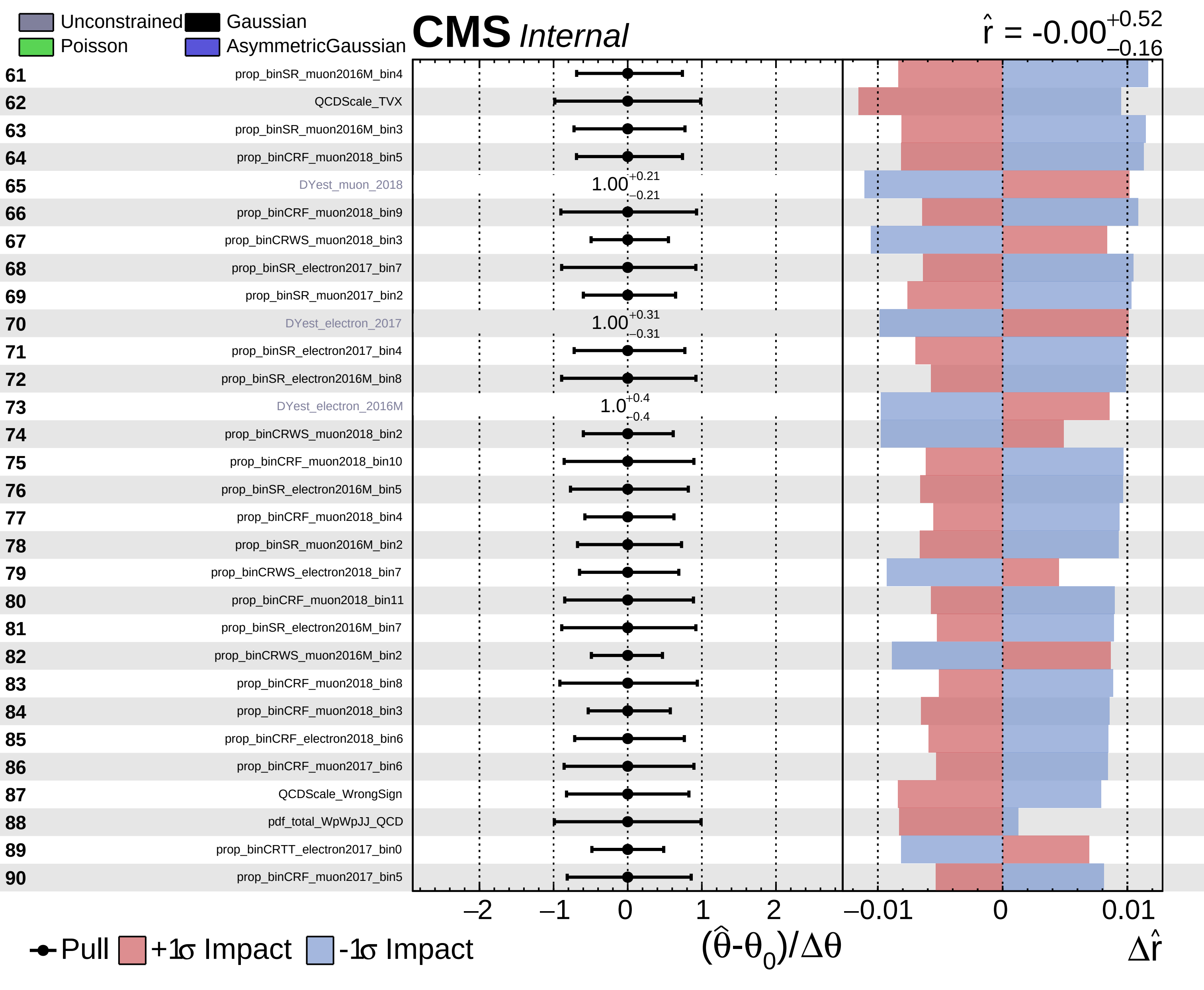
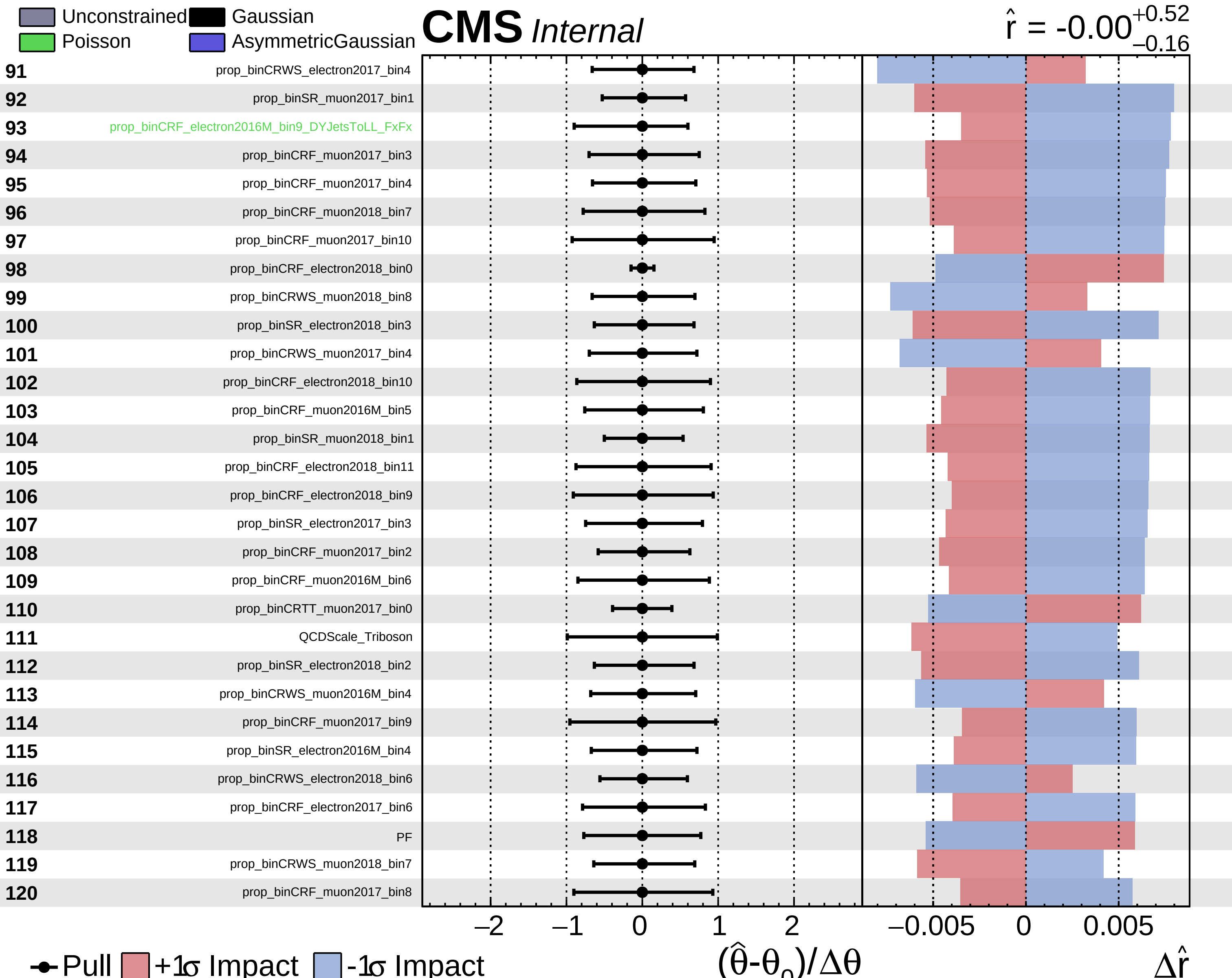

CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2018_bin10
9 prop_binSR_muon2017_bin11
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2018_bin9
12 prop_binSR_muon2016M_bin11
13 prop_binSR_muon2017_bin10
14 prop_binSR_muon2018_bin6
15 prop_binSR_electron2018_bin10
16 prop_binSR_electron2017_bin11
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2016M_bin10
19 prop_binSR_muon2017_bin9
20 prop_binSR_muon2018_bin7
21 prop_binSR_muon2017_bin6
22 prop_binSR_muon2018_bin5
23 prop_binSR_electron2017_bin10
24 prop_binSR_electron2018_bin9
25 prop_binSR_electron2018_bin6
26 lumi_2018
27 prop_binSR_muon2017_bin8
28 prop_binSR_electron2016M_bin11
29 prop_binSR_muon2016M_bin6
30 prop_binSR_muon2016M_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



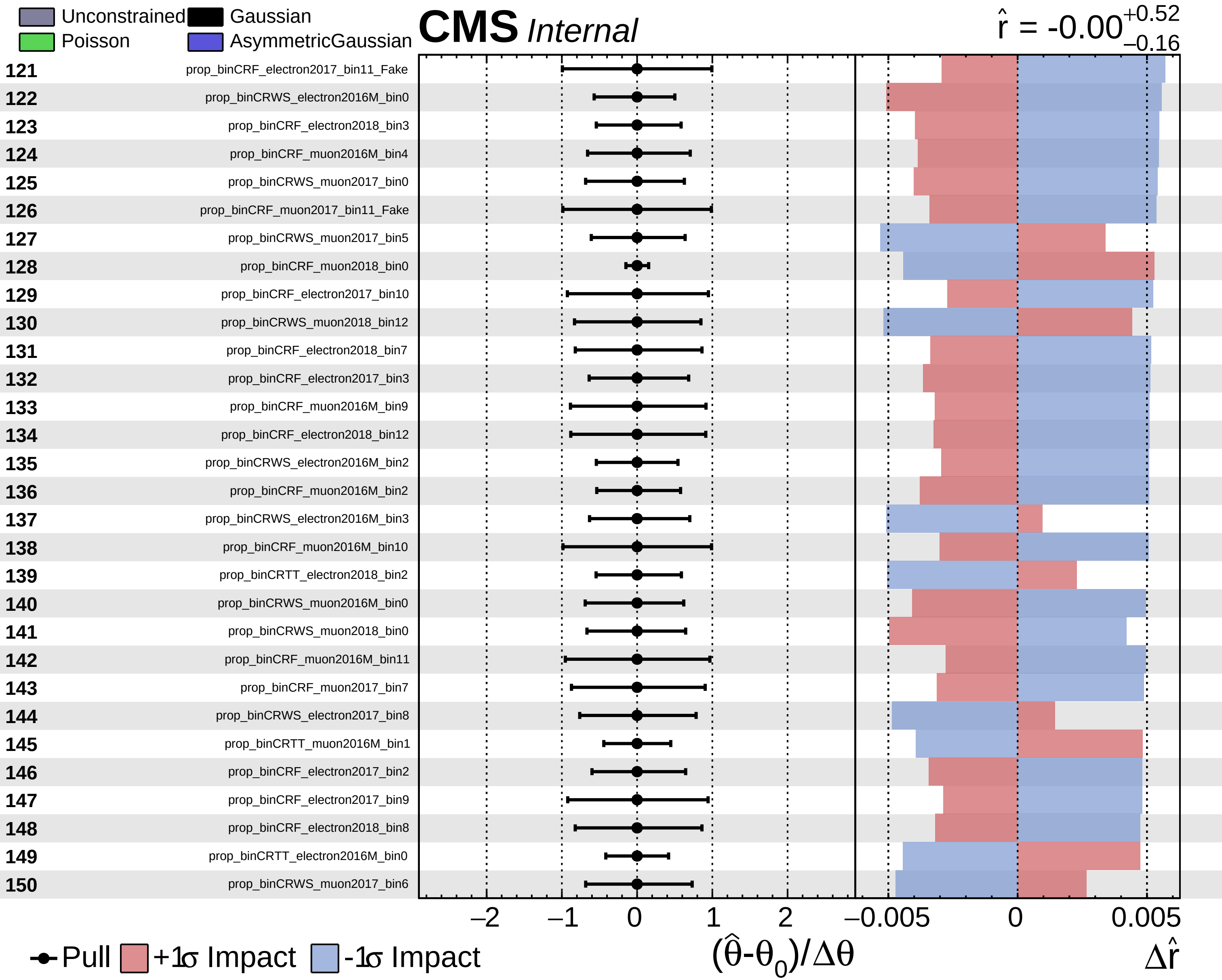
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
31 prop_binSR_muon2018_bin4
32 prop_binSR_electron2018_bin8
33 prop_binSR_muon2018_bin3
34 prop_binSR_electron2017_bin6
35 prop_binSR_electron2016M_bin10
36 prop_binSR_muon2017_bin7
37 lumi_2017
38 prop_binSR_muon2017_bin5
39 prop_binSR_electron2017_bin9
40 prop_binSR_electron2018_bin7
41 prop_binSR_muon2016M_bin8
42 prop_binSR_muon2017_bin4
43 jer_2018
44 prop_binSR_muon2018_bin2
45 prop_binSR_electron2018_bin5
46 QCDScale_WpWpJJ_QCD
47 prop_binSR_muon2016M_bin5
48 lumi_2016M
49 prop_binSR_electron2017_bin8
50 prop_binSR_electron2016M_bin9
51 prop_binSR_muon2016M_bin7
52 DYest_muon_2017
$1.00^{+0.28}_{-0.28}$
53 prop_binSR_electron2018_bin4
54 prop_binSR_electron2017_bin5
55 DYest_electron_2018
$1.0^{+0.4}_{-0.4}$
56 prop_binSR_muon2017_bin3
57 prop_binCRF_muon2018_bin6
58 prop_binCRWS_electron2018_bin5
59 prop_binSR_electron2016M_bin6
60 DYest_muon_2016M
$1.00^{+0.34}_{-0.34}$
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



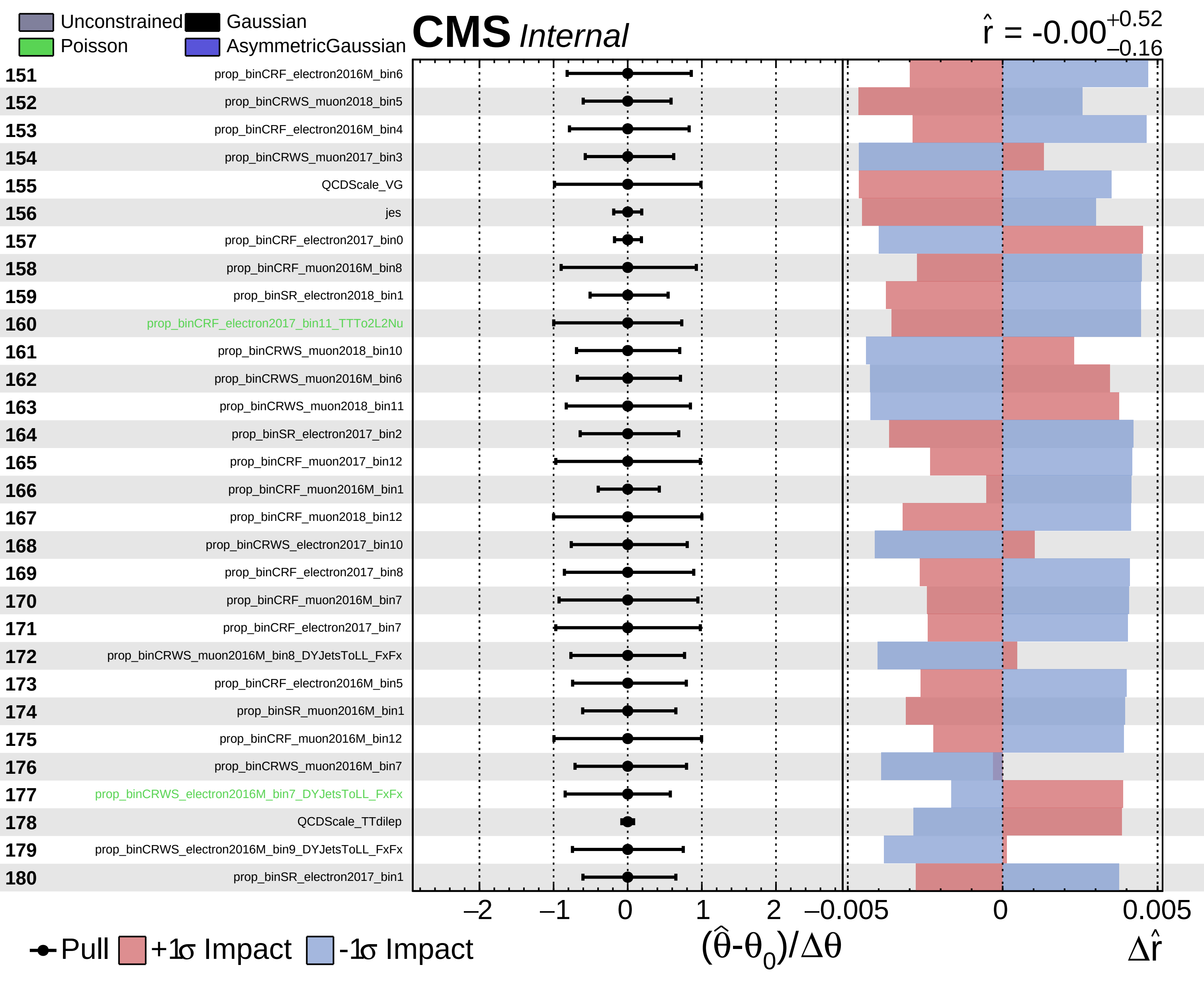
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.52 −0.16
61 prop_binSR_muon2016M_bin4
62 QCDScale_TVX
63 prop_binSR_muon2016M_bin3
64 prop_binCRF_muon2018_bin5
65 DYest_muon_2018 1.00 +0.21 −0.21
66 prop_binCRF_muon2018_bin9
67 prop_binCRWS_muon2018_bin3
68 prop_binSR_electron2017_bin7
69 prop_binSR_muon2017_bin2
70 DYest_electron_2017 1.00 +0.31 −0.31
71 prop_binSR_electron2017_bin4
72 prop_binSR_electron2016M_bin8
73 DYest_electron_2016M 1.0 +0.4 −0.4
74 prop_binCRWS_muon2018_bin2
75 prop_binCRF_muon2018_bin10
76 prop_binSR_electron2016M_bin5
77 prop_binCRF_muon2018_bin4
78 prop_binSR_muon2016M_bin2
79 prop_binCRWS_electron2018_bin7
80 prop_binCRF_muon2018_bin11
81 prop_binSR_electron2016M_bin7
82 prop_binCRWS_muon2016M_bin2
83 prop_binCRF_muon2018_bin8
84 prop_binCRF_muon2018_bin3
85 prop_binCRF_electron2018_bin6
86 prop_binCRF_muon2017_bin6
87 QCDScale_WrongSign
88 pdf_total_WpWpJJ_QCD
89 prop_binCRTT_electron2017_bin0
90 prop_binCRF_muon2017_bin5
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

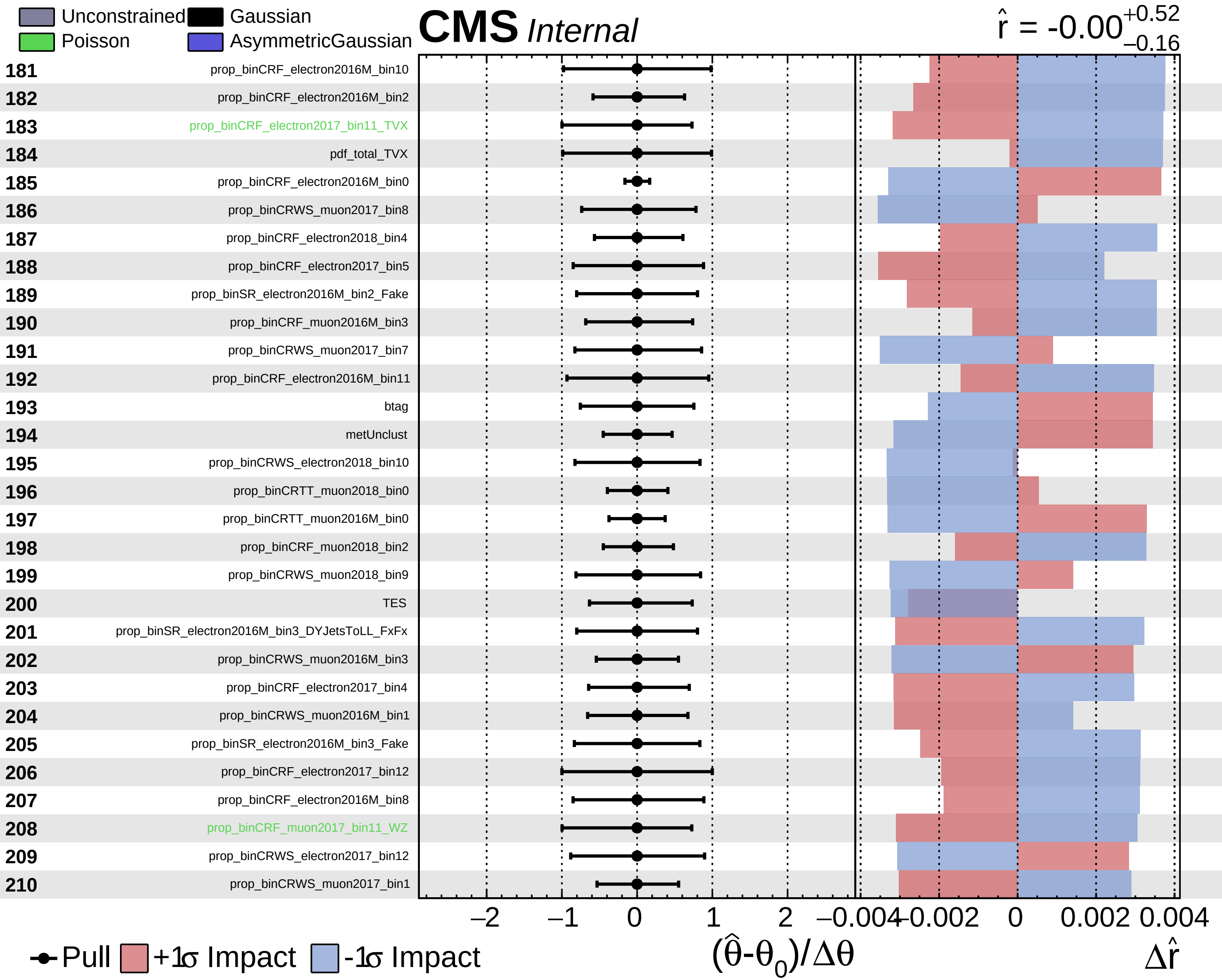


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
91 prop_binCRWS_electron2017_bin4
92 prop_binSR_muon2017_bin1
93 prop_binCRF_electron2016M_bin9_DYJetsToLL_FxFx
94 prop_binCRF_muon2017_bin3
95 prop_binCRF_muon2017_bin4
96 prop_binCRF_muon2018_bin7
97 prop_binCRF_muon2017_bin10
98 prop_binCRF_electron2018_bin0
99 prop_binCRWS_muon2018_bin8
100 prop_binSR_electron2018_bin3
101 prop_binCRWS_muon2017_bin4
102 prop_binCRF_electron2018_bin10
103 prop_binCRF_muon2016M_bin5
104 prop_binSR_muon2018_bin1
105 prop_binCRF_electron2018_bin11
106 prop_binCRF_electron2018_bin9
107 prop_binSR_electron2017_bin3
108 prop_binCRF_muon2017_bin2
109 prop_binCRF_muon2016M_bin6
110 prop_binCRTT_muon2017_bin0
111 QCDScale_Triboson
112 prop_binSR_electron2018_bin2
113 prop_binCRWS_muon2016M_bin4
114 prop_binCRF_muon2017_bin9
115 prop_binSR_electron2016M_bin4
116 prop_binCRWS_electron2018_bin6
117 prop_binCRF_electron2017_bin6
118 PF
119 prop_binCRWS_muon2018_bin7
120 prop_binCRF_muon2017_bin8
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

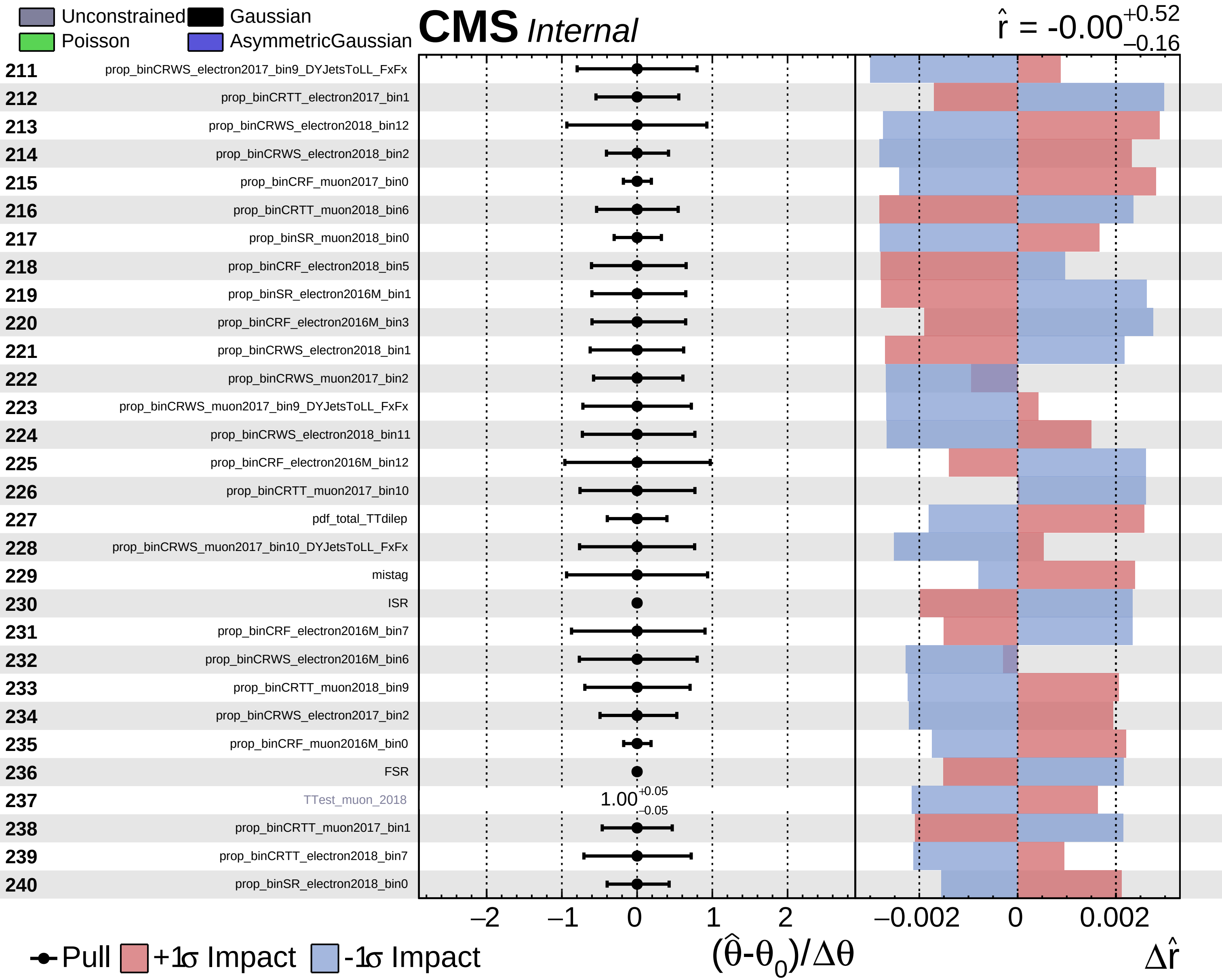


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
121 prop_binCRF_electron2017_bin11_Fake
122 prop_binCRWS_electron2016M_bin0
123 prop_binCRF_electron2018_bin3
124 prop_binCRF_muon2016M_bin4
125 prop_binCRWS_muon2017_bin0
126 prop_binCRF_muon2017_bin11_Fake
127 prop_binCRWS_muon2017_bin5
128 prop_binCRF_muon2018_bin0
129 prop_binCRF_electron2017_bin10
130 prop_binCRWS_muon2018_bin12
131 prop_binCRF_electron2018_bin7
132 prop_binCRF_electron2017_bin3
133 prop_binCRF_muon2016M_bin9
134 prop_binCRF_electron2018_bin12
135 prop_binCRWS_electron2016M_bin2
136 prop_binCRF_muon2016M_bin2
137 prop_binCRWS_electron2016M_bin3
138 prop_binCRF_muon2016M_bin10
139 prop_binCRTT_electron2018_bin2
140 prop_binCRWS_muon2016M_bin0
141 prop_binCRWS_muon2018_bin0
142 prop_binCRF_muon2016M_bin11
143 prop_binCRF_muon2017_bin7
144 prop_binCRWS_electron2017_bin8
145 prop_binCRTT_muon2016M_bin1
146 prop_binCRF_electron2017_bin2
147 prop_binCRF_electron2017_bin9
148 prop_binCRF_electron2018_bin8
149 prop_binCRTT_electron2016M_bin0
150 prop_binCRWS_muon2017_bin6
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

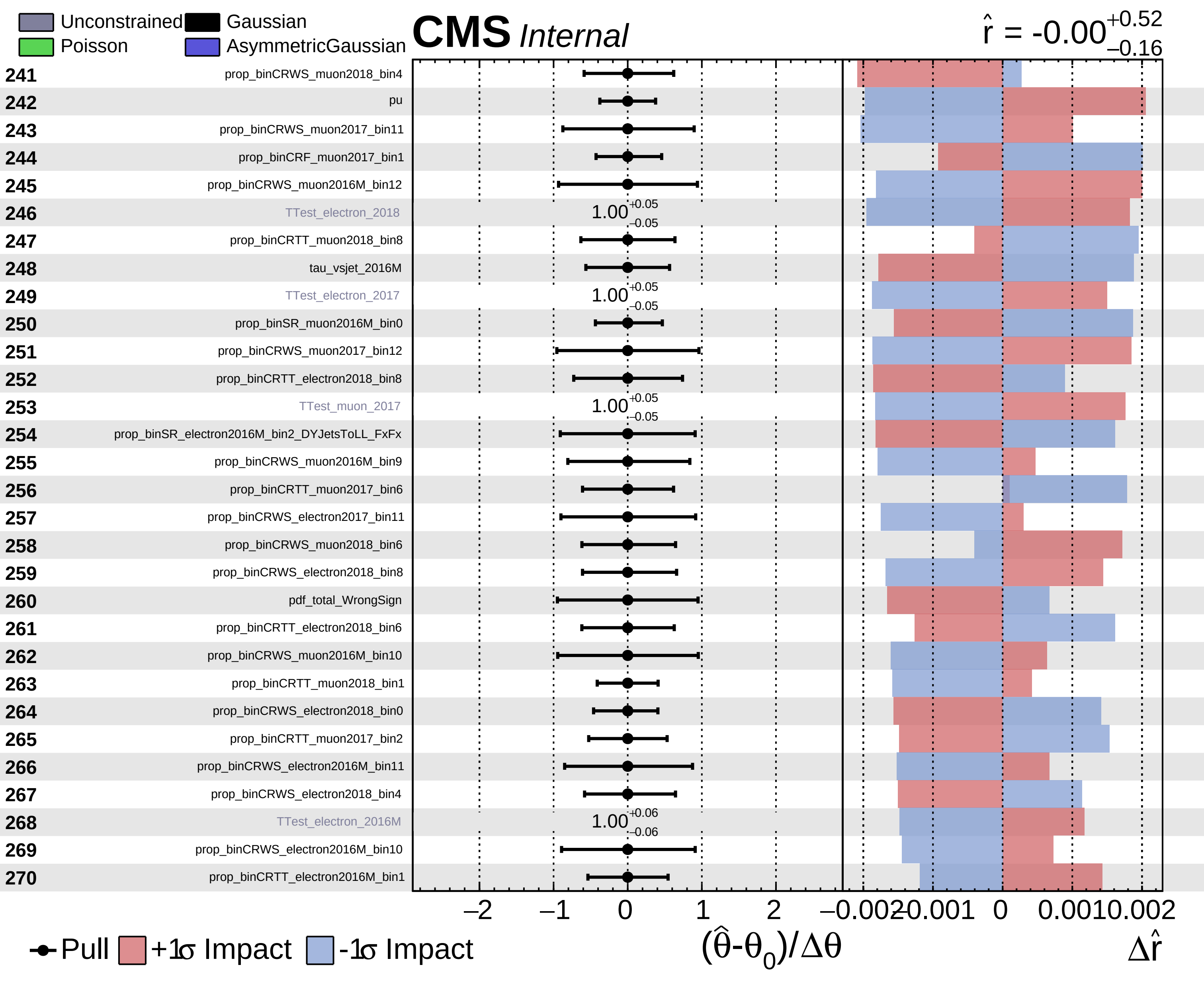

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
151 prop_binCRF_electron2016M_bin6
152 prop_binCRWS_muon2018_bin5
153 prop_binCRF_electron2016M_bin4
154 prop_binCRWS_muon2017_bin3
155 QCDScale_VG
156 jes
157 prop_binCRF_electron2017_bin0
158 prop_binCRF_muon2016M_bin8
159 prop_binSR_electron2018_bin1
160 prop_binCRF_electron2017_bin11_TTTo2L2Nu
161 prop_binCRWS_muon2018_bin10
162 prop_binCRWS_muon2016M_bin6
163 prop_binCRWS_muon2018_bin11
164 prop_binSR_electron2017_bin2
165 prop_binCRF_muon2017_bin12
166 prop_binCRF_muon2016M_bin1
167 prop_binCRF_muon2018_bin12
168 prop_binCRWS_electron2017_bin10
169 prop_binCRF_electron2017_bin8
170 prop_binCRF_muon2016M_bin7
171 prop_binCRF_electron2017_bin7
172 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
173 prop_binCRF_electron2016M_bin5
174 prop_binSR_muon2016M_bin1
175 prop_binCRF_muon2016M_bin12
176 prop_binCRWS_muon2016M_bin7
177 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
178 QCDScale_TTdilep
179 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
180 prop_binSR_electron2017_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



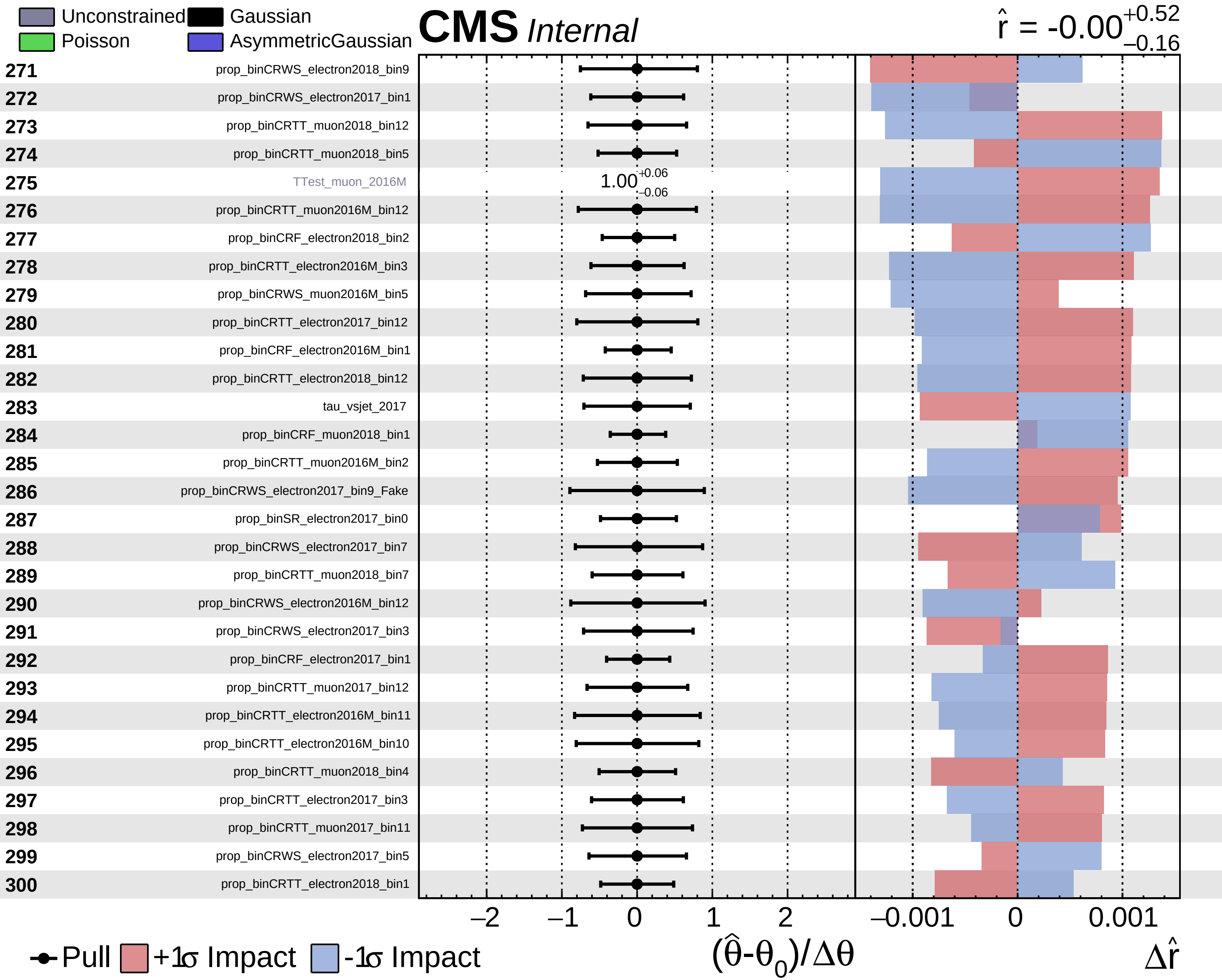
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
181 prop_binCRF_electron2016M_bin10
182 prop_binCRF_electron2016M_bin2
183 prop_binCRF_electron2017_bin11_TVX
184 pdf_total_TVX
185 prop_binCRF_electron2016M_bin0
186 prop_binCRWS_muon2017_bin8
187 prop_binCRF_electron2018_bin4
188 prop_binCRF_electron2017_bin5
189 prop_binSR_electron2016M_bin2_Fake
190 prop_binCRF_muon2016M_bin3
191 prop_binCRWS_muon2017_bin7
192 prop_binCRF_electron2016M_bin11
193 btag
194 metUnclust
195 prop_binCRWS_electron2018_bin10
196 prop_binCRTT_muon2018_bin0
197 prop_binCRTT_muon2016M_bin0
198 prop_binCRF_muon2018_bin2
199 prop_binCRWS_muon2018_bin9
200 TES
201 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
202 prop_binCRWS_muon2016M_bin3
203 prop_binCRF_electron2017_bin4
204 prop_binCRWS_muon2016M_bin1
205 prop_binSR_electron2016M_bin3_Fake
206 prop_binCRF_electron2017_bin12
207 prop_binCRF_electron2016M_bin8
208 prop_binCRF_muon2017_bin11_WZ
209 prop_binCRWS_electron2017_bin12
210 prop_binCRWS_muon2017_bin1
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

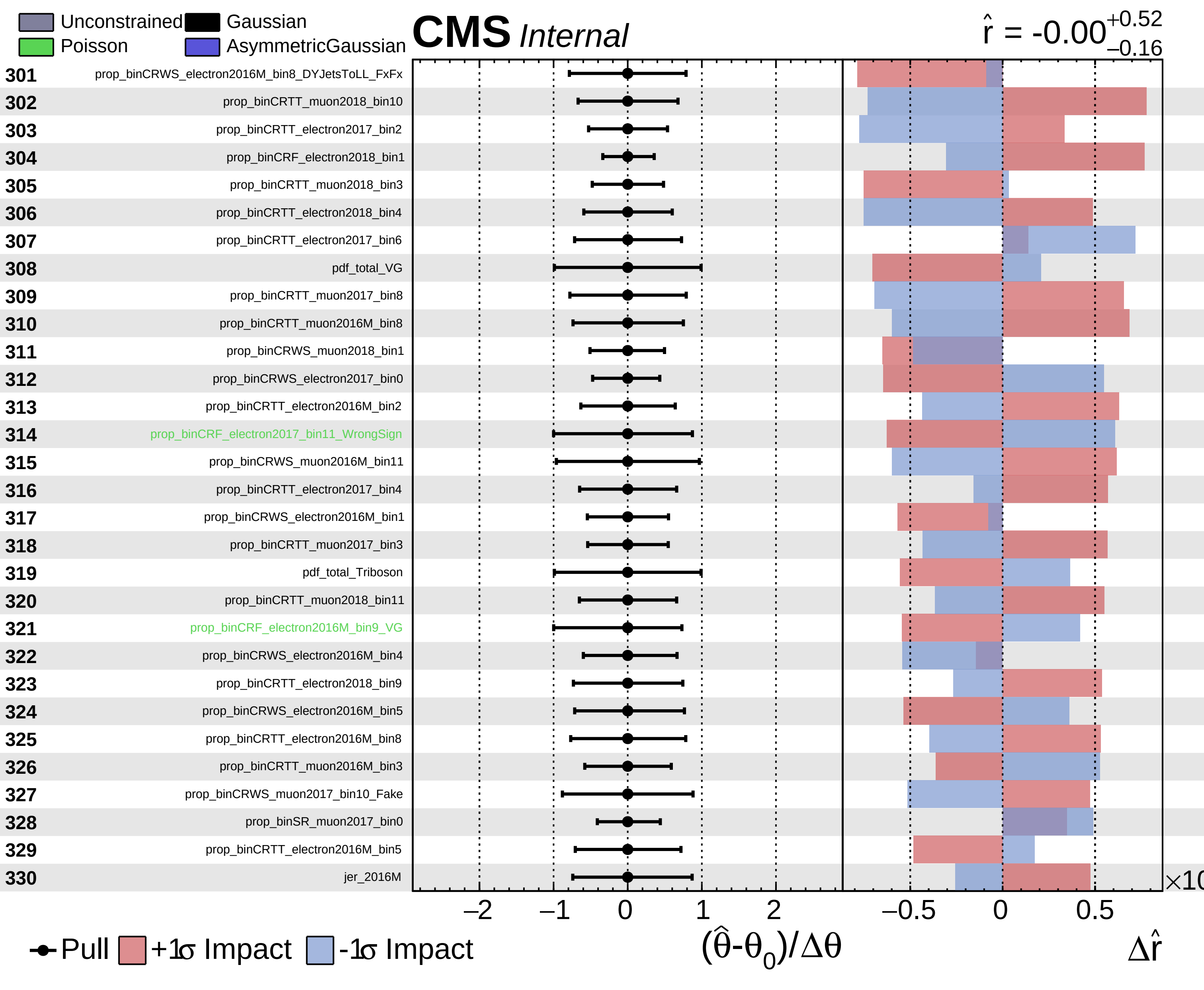


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
211 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
212 prop_binCRTT_electron2017_bin1
213 prop_binCRWS_electron2018_bin12
214 prop_binCRWS_electron2018_bin2
215 prop_binCRF_muon2017_bin0
216 prop_binCRTT_muon2018_bin6
217 prop_binSR_muon2018_bin0
218 prop_binCRF_electron2018_bin5
219 prop_binSR_electron2016M_bin1
220 prop_binCRF_electron2016M_bin3
221 prop_binCRWS_electron2018_bin1
222 prop_binCRWS_muon2017_bin2
223 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
224 prop_binCRWS_electron2018_bin11
225 prop_binCRF_electron2016M_bin12
226 prop_binCRTT_muon2017_bin10
227 pdf_total_TTdilep
228 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
229 mistag
230 ISR
231 prop_binCRF_electron2016M_bin7
232 prop_binCRWS_electron2016M_bin6
233 prop_binCRTT_muon2018_bin9
234 prop_binCRWS_electron2017_bin2
235 prop_binCRF_muon2016M_bin0
236 FSR
237 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
238 prop_binCRTT_muon2017_bin1
239 prop_binCRTT_electron2018_bin7
240 prop_binSR_electron2018_bin0
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

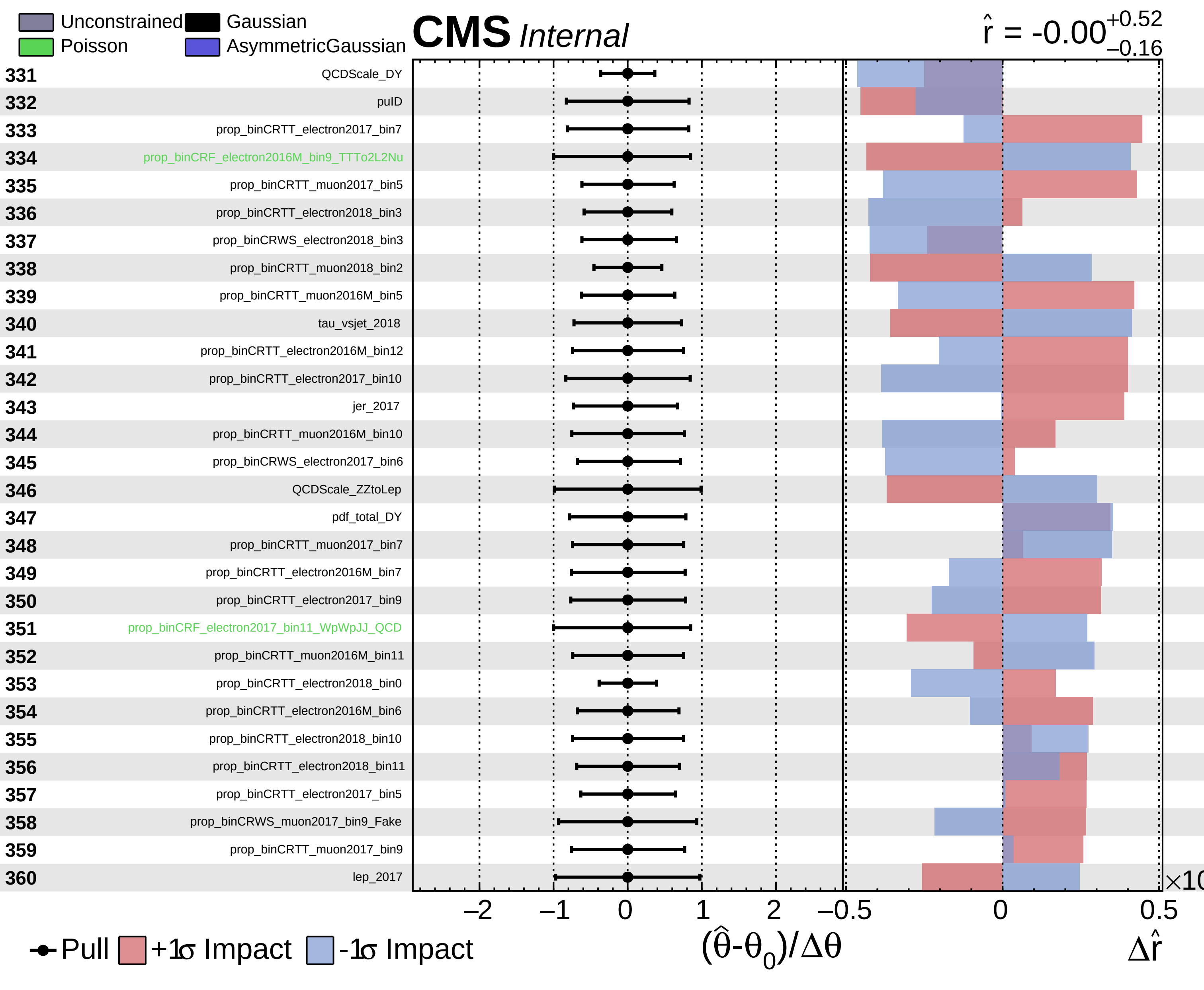


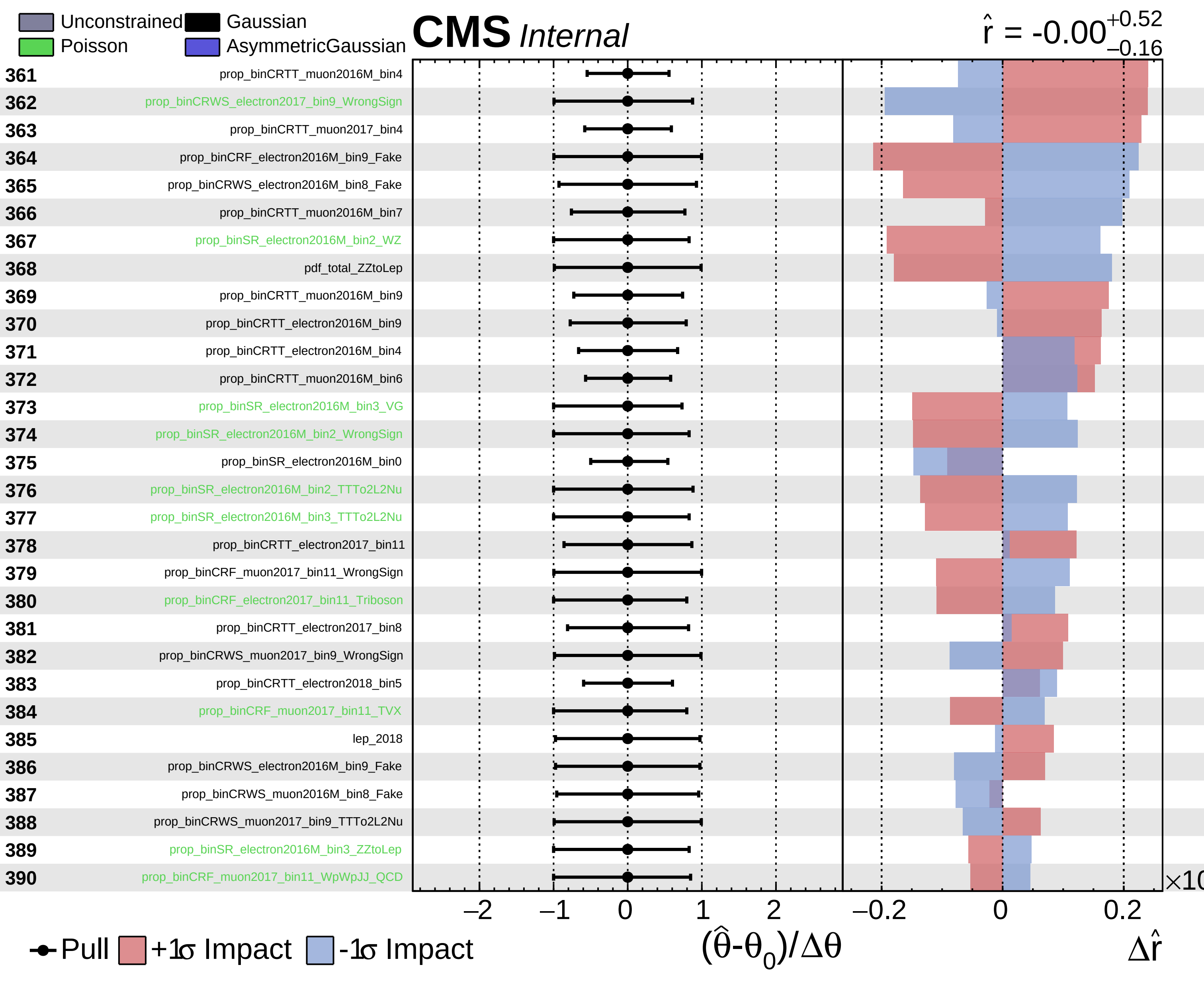
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.52 −0.16
241 prop_binCRWS_muon2018_bin4
242 pu
243 prop_binCRWS_muon2017_bin11
244 prop_binCRF_muon2017_bin1
245 prop_binCRWS_muon2016M_bin12
246 TTest_electron_2018 1.00 +0.05 −0.05
247 prop_binCRTT_muon2018_bin8
248 tau_vsjet_2016M
249 TTest_electron_2017 1.00 +0.05 −0.05
250 prop_binSR_muon2016M_bin0
251 prop_binCRWS_muon2017_bin12
252 prop_binCRTT_electron2018_bin8
253 TTest_muon_2017 1.00 +0.05 −0.05
254 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
255 prop_binCRWS_muon2016M_bin9
256 prop_binCRTT_muon2017_bin6
257 prop_binCRWS_electron2017_bin11
258 prop_binCRWS_muon2018_bin6
259 prop_binCRWS_electron2018_bin8
260 pdf_total_WrongSign
261 prop_binCRTT_electron2018_bin6
262 prop_binCRWS_muon2016M_bin10
263 prop_binCRTT_muon2018_bin1
264 prop_binCRWS_electron2018_bin0
265 prop_binCRTT_muon2017_bin2
266 prop_binCRWS_electron2016M_bin11
267 prop_binCRWS_electron2018_bin4
268 TTest_electron_2016M 1.00 +0.06 −0.06
269 prop_binCRWS_electron2016M_bin10
270 prop_binCRTT_electron2016M_bin1
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

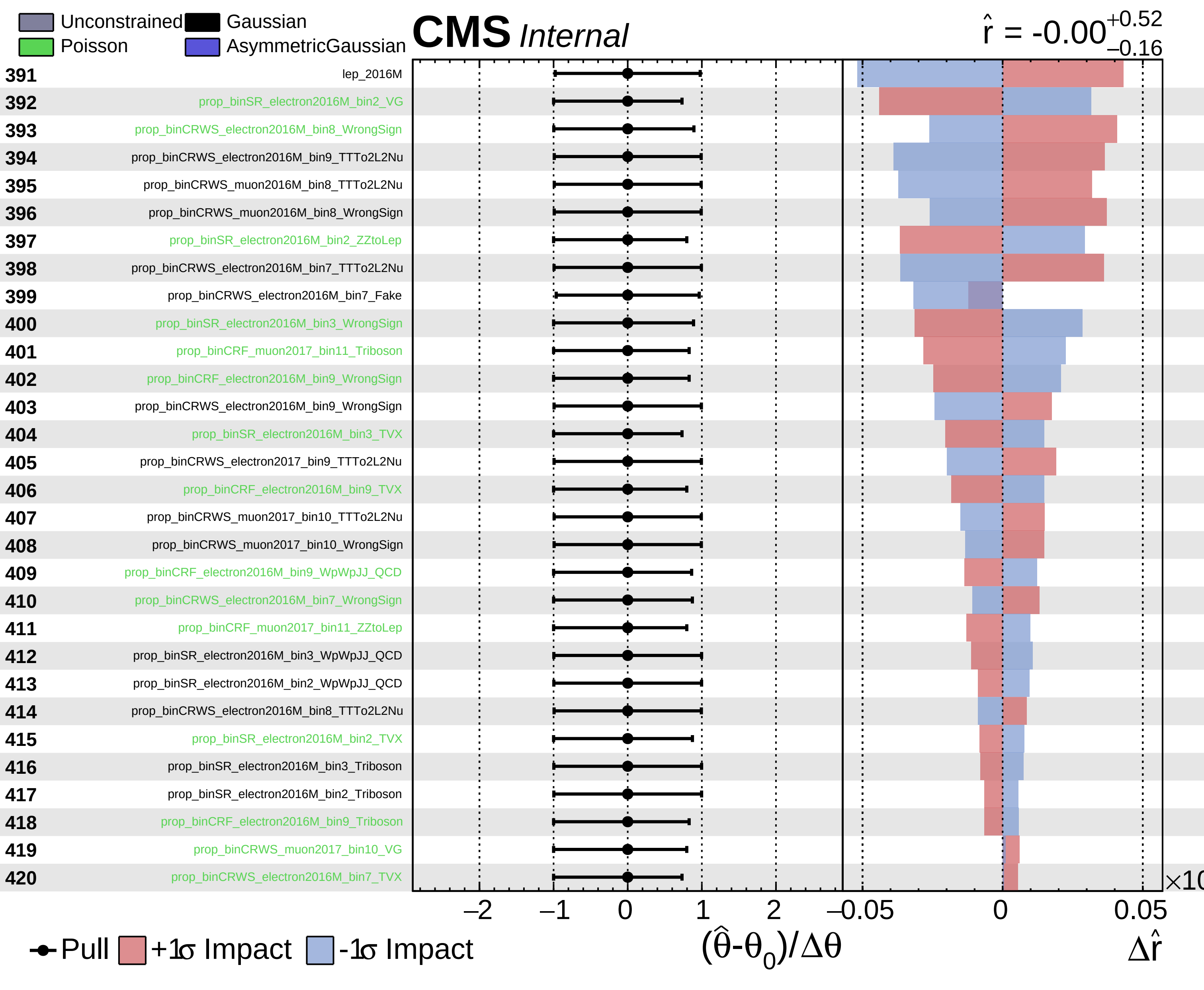

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
271 prop_binCRWS_electron2018_bin9
272 prop_binCRWS_electron2017_bin1
273 prop_binCRTT_muon2018_bin12
274 prop_binCRTT_muon2018_bin5
275 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
276 prop_binCRTT_muon2016M_bin12
277 prop_binCRF_electron2018_bin2
278 prop_binCRTT_electron2016M_bin3
279 prop_binCRWS_muon2016M_bin5
280 prop_binCRTT_electron2017_bin12
281 prop_binCRF_electron2016M_bin1
282 prop_binCRTT_electron2018_bin12
283 tau_vsjet_2017
284 prop_binCRF_muon2018_bin1
285 prop_binCRTT_muon2016M_bin2
286 prop_binCRWS_electron2017_bin9_Fake
287 prop_binSR_electron2017_bin0
288 prop_binCRWS_electron2017_bin7
289 prop_binCRTT_muon2018_bin7
290 prop_binCRWS_electron2016M_bin12
291 prop_binCRWS_electron2017_bin3
292 prop_binCRF_electron2017_bin1
293 prop_binCRTT_muon2017_bin12
294 prop_binCRTT_electron2016M_bin11
295 prop_binCRTT_electron2016M_bin10
296 prop_binCRTT_muon2018_bin4
297 prop_binCRTT_electron2017_bin3
298 prop_binCRTT_muon2017_bin11
299 prop_binCRWS_electron2017_bin5
300 prop_binCRTT_electron2018_bin1
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



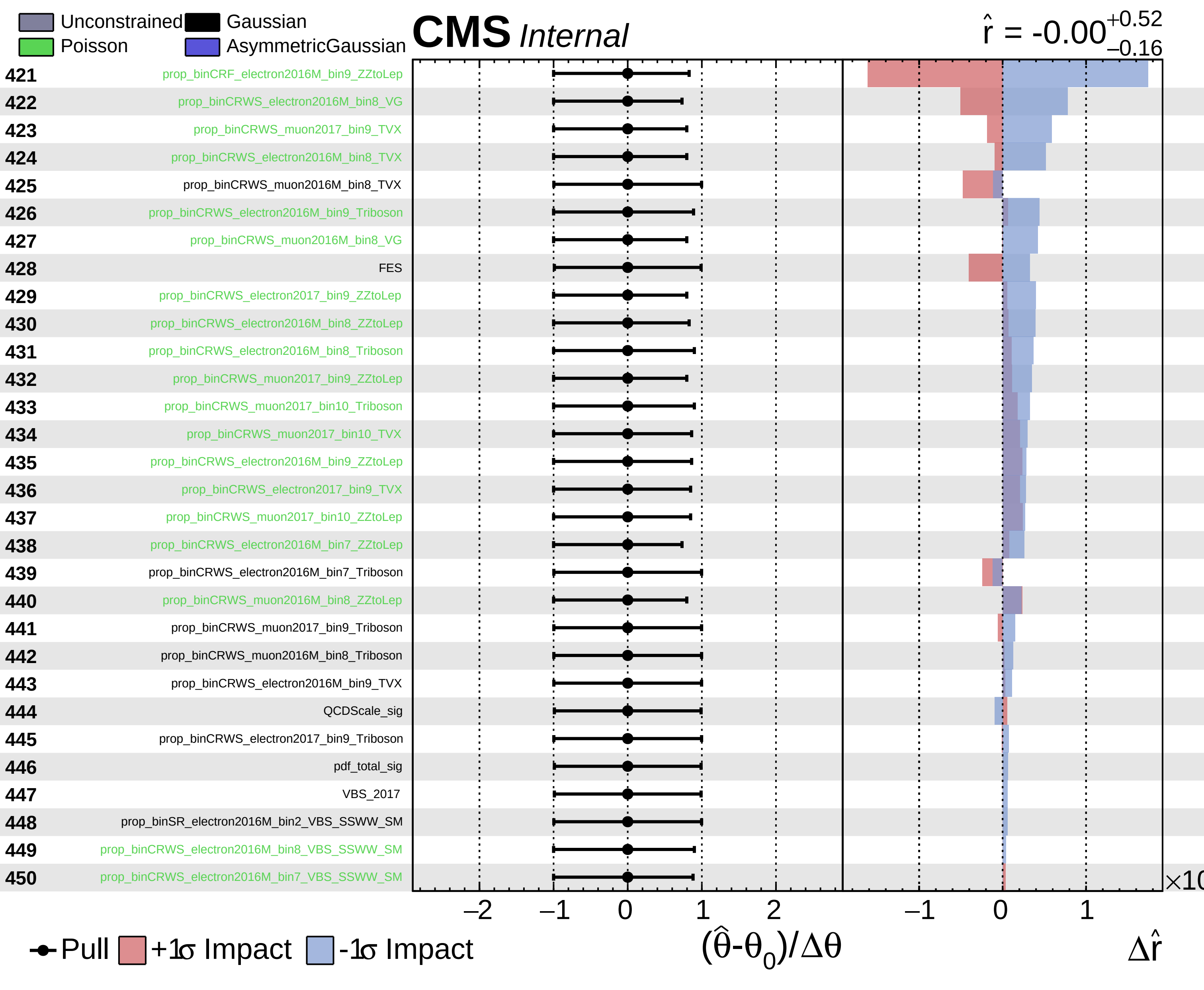
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.52 −0.16
301 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
302 prop_binCRTT_muon2018_bin10
303 prop_binCRTT_electron2017_bin2
304 prop_binCRF_electron2018_bin1
305 prop_binCRTT_muon2018_bin3
306 prop_binCRTT_electron2018_bin4
307 prop_binCRTT_electron2017_bin6
308 pdf_total_VG
309 prop_binCRTT_muon2017_bin8
310 prop_binCRTT_muon2016M_bin8
311 prop_binCRWS_muon2018_bin1
312 prop_binCRWS_electron2017_bin0
313 prop_binCRTT_electron2016M_bin2
314 prop_binCRF_electron2017_bin11_WrongSign
315 prop_binCRWS_muon2016M_bin11
316 prop_binCRTT_electron2017_bin4
317 prop_binCRWS_electron2016M_bin1
318 prop_binCRTT_muon2017_bin3
319 pdf_total_Triboson
320 prop_binCRTT_muon2018_bin11
321 prop_binCRF_electron2016M_bin9_VG
322 prop_binCRWS_electron2016M_bin4
323 prop_binCRTT_electron2018_bin9
324 prop_binCRWS_electron2016M_bin5
325 prop_binCRTT_electron2016M_bin8
326 prop_binCRTT_muon2016M_bin3
327 prop_binCRWS_muon2017_bin10_Fake
328 prop_binSR_muon2017_bin0
329 prop_binCRTT_electron2016M_bin5
330 jer_2016M
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

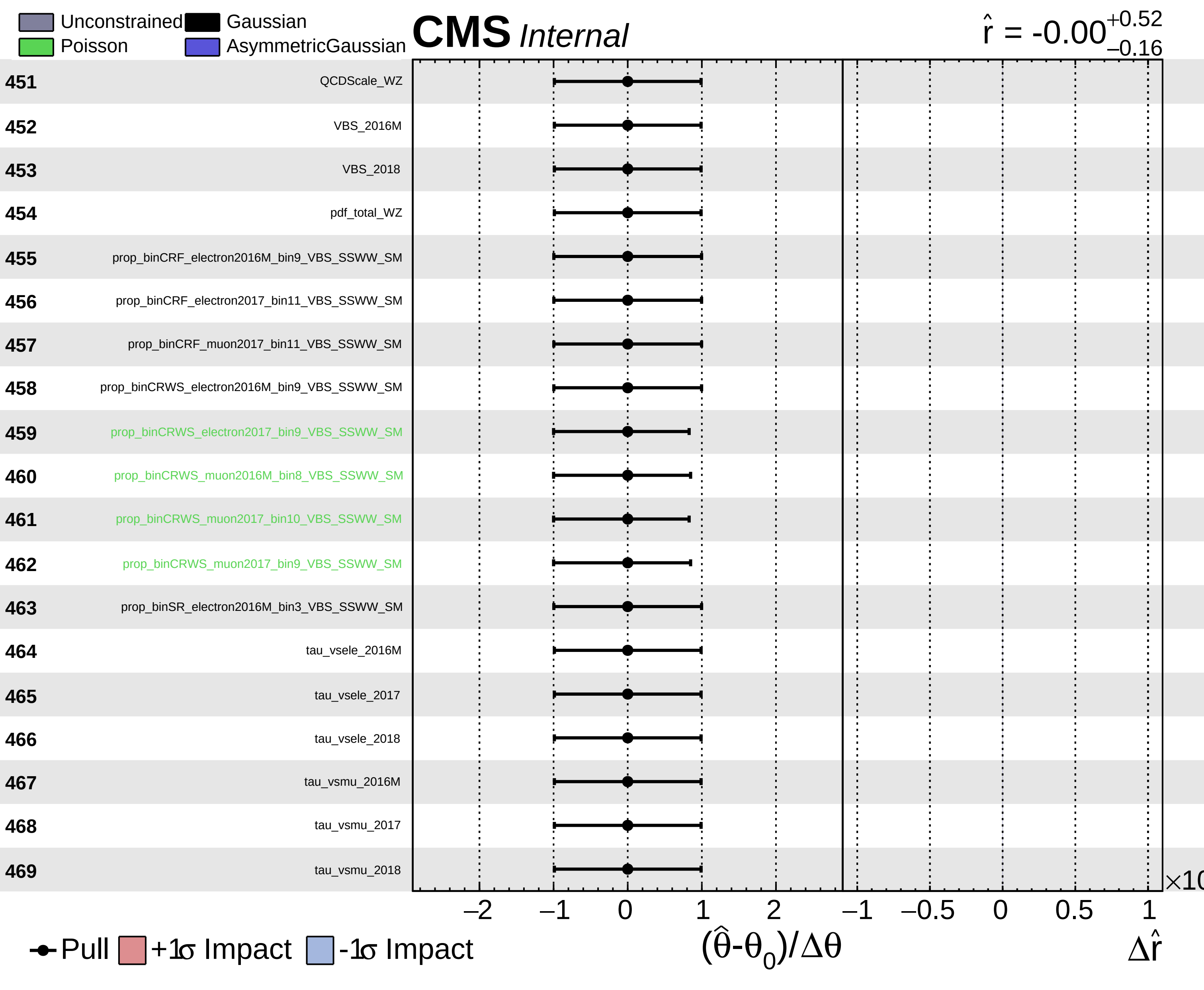

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
331 QCDScale_DY
332 puID
333 prop_binCRTT_electron2017_bin7
334 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
335 prop_binCRTT_muon2017_bin5
336 prop_binCRTT_electron2018_bin3
337 prop_binCRWS_electron2018_bin3
338 prop_binCRTT_muon2018_bin2
339 prop_binCRTT_muon2016M_bin5
340 tau_vsjet_2018
341 prop_binCRTT_electron2016M_bin12
342 prop_binCRTT_electron2017_bin10
343 jer_2017
344 prop_binCRTT_muon2016M_bin10
345 prop_binCRWS_electron2017_bin6
346 QCDScale_ZZtoLep
347 pdf_total_DY
348 prop_binCRTT_muon2017_bin7
349 prop_binCRTT_electron2016M_bin7
350 prop_binCRTT_electron2017_bin9
351 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
352 prop_binCRTT_muon2016M_bin11
353 prop_binCRTT_electron2018_bin0
354 prop_binCRTT_electron2016M_bin6
355 prop_binCRTT_electron2018_bin10
356 prop_binCRTT_electron2018_bin11
357 prop_binCRTT_electron2017_bin5
358 prop_binCRWS_muon2017_bin9_Fake
359 prop_binCRTT_muon2017_bin9
360 lep_2017
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
361 prop_binCRTT_muon2016M_bin4
362 prop_binCRWS_electron2017_bin9_WrongSign
363 prop_binCRTT_muon2017_bin4
364 prop_binCRF_electron2016M_bin9_Fake
365 prop_binCRWS_electron2016M_bin8_Fake
366 prop_binCRTT_muon2016M_bin7
367 prop_binSR_electron2016M_bin2_WZ
368 pdf_total_ZZtoLep
369 prop_binCRTT_muon2016M_bin9
370 prop_binCRTT_electron2016M_bin9
371 prop_binCRTT_electron2016M_bin4
372 prop_binCRTT_muon2016M_bin6
373 prop_binSR_electron2016M_bin3_VG
374 prop_binSR_electron2016M_bin2_WrongSign
375 prop_binSR_electron2016M_bin0
376 prop_binSR_electron2016M_bin2_TTTo2L2Nu
377 prop_binSR_electron2016M_bin3_TTTo2L2Nu
378 prop_binCRTT_electron2017_bin11
379 prop_binCRF_muon2017_bin11_WrongSign
380 prop_binCRF_electron2017_bin11_Triboson
381 prop_binCRTT_electron2017_bin8
382 prop_binCRWS_muon2017_bin9_WrongSign
383 prop_binCRTT_electron2018_bin5
384 prop_binCRF_muon2017_bin11_TVX
385 lep_2018
386 prop_binCRWS_electron2016M_bin9_Fake
387 prop_binCRWS_muon2016M_bin8_Fake
388 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
389 prop_binSR_electron2016M_bin3_ZZtoLep
390 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
391 lep_2016M
392 prop_binSR_electron2016M_bin2_VG
393 prop_binCRWS_electron2016M_bin8_WrongSign
394 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
395 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
396 prop_binCRWS_muon2016M_bin8_WrongSign
397 prop_binSR_electron2016M_bin2_ZZtoLep
398 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
399 prop_binCRWS_electron2016M_bin7_Fake
400 prop_binSR_electron2016M_bin3_WrongSign
401 prop_binCRF_muon2017_bin11_Triboson
402 prop_binCRF_electron2016M_bin9_WrongSign
403 prop_binCRWS_electron2016M_bin9_WrongSign
404 prop_binSR_electron2016M_bin3_TVX
405 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
406 prop_binCRF_electron2016M_bin9_TVX
407 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
408 prop_binCRWS_muon2017_bin10_WrongSign
409 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
410 prop_binCRWS_electron2016M_bin7_WrongSign
411 prop_binCRF_muon2017_bin11_ZZtoLep
412 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
413 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
414 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
415 prop_binSR_electron2016M_bin2_TVX
416 prop_binSR_electron2016M_bin3_Triboson
417 prop_binSR_electron2016M_bin2_Triboson
418 prop_binCRF_electron2016M_bin9_Triboson
419 prop_binCRWS_muon2017_bin10_VG
420 prop_binCRWS_electron2016M_bin7_TVX
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.52 −0.16
421 prop_binCRF_electron2016M_bin9_ZZtoLep
422 prop_binCRWS_electron2016M_bin8_VG
423 prop_binCRWS_muon2017_bin9_TVX
424 prop_binCRWS_electron2016M_bin8_TVX
425 prop_binCRWS_muon2016M_bin8_TVX
426 prop_binCRWS_electron2016M_bin9_Triboson
427 prop_binCRWS_muon2016M_bin8_VG
428 FES
429 prop_binCRWS_electron2017_bin9_ZZtoLep
430 prop_binCRWS_electron2016M_bin8_ZZtoLep
431 prop_binCRWS_electron2016M_bin8_Triboson
432 prop_binCRWS_muon2017_bin9_ZZtoLep
433 prop_binCRWS_muon2017_bin10_Triboson
434 prop_binCRWS_muon2017_bin10_TVX
435 prop_binCRWS_electron2016M_bin9_ZZtoLep
436 prop_binCRWS_electron2017_bin9_TVX
437 prop_binCRWS_muon2017_bin10_ZZtoLep
438 prop_binCRWS_electron2016M_bin7_ZZtoLep
439 prop_binCRWS_electron2016M_bin7_Triboson
440 prop_binCRWS_muon2016M_bin8_ZZtoLep
441 prop_binCRWS_muon2017_bin9_Triboson
442 prop_binCRWS_muon2016M_bin8_Triboson
443 prop_binCRWS_electron2016M_bin9_TVX
444 QCDScale_sig
445 prop_binCRWS_electron2017_bin9_Triboson
446 pdf_total_sig
447 VBS_2017
448 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
449 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
450 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.52}_{-0.16}$
451 QCDScale_WZ
452 VBS_2016M
453 VBS_2018
454 pdf_total_WZ
455 prop_binCRF_electron2016M_bin9_VBS_SSWW_SM
456 prop_binCRF_electron2017_bin11_VBS_SSWW_SM
457 prop_binCRF_muon2017_bin11_VBS_SSWW_SM
458 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
459 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
460 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
461 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
462 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
463 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
464 tau_vsele_2016M
465 tau_vsele_2017
466 tau_vsele_2018
467 tau_vsmu_2016M
468 tau_vsmu_2017
469 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$